

HELMCKEN ISLAND, BC, Canada

WMO: 714813

Lat: 50.400N

Lon: 125.870W

Elev: 62

StdP: 14.66

Time Zone: -8.00 (NAP)

Period: 90-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.4	28.4	8.6	8.5	29.8	14.4	11.4	30.6	36.6	44.4	35.3	44.0	10.6	90	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.1	71.1	57.4	68.4	56.6	66.2	55.7	60.0	66.9	58.7	64.8	57.5	63.1	6.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.5	70.8	60.8	56.1	67.3	59.6	55.0	64.5	58.8	26.3	66.6	25.5	65.1	24.7	63.6	66.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
33.5	28.7	24.3	23.7	79.4	4.8	3.0	20.3	81.5	17.5	83.3	14.8	85.0	11.4	87.2		
			DB													
			WB	20.0	62.2	4.8	2.5	16.6	64.1	13.8	65.6	11.1	67.0	7.7	68.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	38.9	41.0	43.3	47.6	52.5	55.8	59.6	59.7	55.8	48.6	42.9	39.0
	DBStd	8.43	5.09	4.07	3.66	3.56	4.08	3.36	2.86	3.14	3.02	3.59	4.28	4.97
	HDD50	1540	345	253	208	89	20	1	0	0	1	68	214	341
	HDD65	5927	810	673	672	522	387	275	168	167	275	509	663	806
	CDD50	1093	0	0	1	17	98	176	298	301	176	25	1	0
	CDD65	4	0	0	0	0	0	0	1	3	0	0	0	0
	CDH74	30	0	0	0	0	3	2	11	14	0	0	0	0
	CDH80	2	0	0	0	0	0	0	1	1	0	0	0	0
Wind	WSAvg	8.3	10.1	10.0	9.1	8.4	7.2	6.5	6.4	5.9	6.1	8.3	10.1	11.7
Precipitation	PrecAvg	82.3	10.8	6.5	6.7	5.8	4.0	3.6	2.5	2.9	4.2	10.0	12.6	11.8
	PrecMax	102.3	17.1	16.8	12.0	13.1	8.5	7.6	5.1	6.5	10.8	15.4	20.2	17.3
	PrecMin	62.1	5.6	0.1	0.7	1.6	1.7	1.5	0.3	0.2	0.8	2.1	5.6	5.9
	PrecStd	10.4	3.3	3.7	2.6	3.1	1.6	1.6	1.4	1.9	2.3	4.0	4.2	2.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.4	51.6	56.5	63.8	71.7	72.6	74.8	75.8	68.8	61.0	53.0	48.9
		MCWB	46.6	45.5	47.1	50.1	54.4	57.4	59.3	59.6	56.3	54.1	50.3	44.6
	2%	DB	47.6	48.9	52.2	58.7	65.7	67.7	70.5	70.7	65.5	56.9	51.0	46.8
		MCWB	44.9	44.8	45.0	48.2	53.0	55.5	57.8	57.9	55.6	51.0	48.0	43.5
	5%	DB	46.0	47.6	50.1	55.5	61.9	64.5	67.8	68.0	63.0	54.8	49.4	45.8
		MCWB	43.7	44.1	44.3	46.6	51.6	54.3	56.7	56.8	54.6	50.1	46.6	43.0
	10%	DB	44.9	46.2	48.5	53.1	59.1	61.9	65.6	65.6	60.9	53.0	48.1	44.6
		MCWB	43.0	43.3	44.2	46.2	50.2	52.7	55.6	56.0	53.7	49.1	45.6	42.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.8	48.4	49.1	52.0	56.8	59.4	61.5	62.9	59.9	55.8	51.4	46.6
		MCDB	48.8	50.3	52.8	58.4	65.4	69.0	72.0	69.8	63.9	58.5	53.0	47.8
	2%	WB	45.5	46.3	47.6	50.6	54.9	57.3	59.4	60.3	57.9	52.7	48.8	44.5
		MCDB	46.9	47.5	50.1	55.3	62.2	64.2	67.3	66.3	62.0	55.4	50.5	45.8
	5%	WB	44.4	45.1	46.6	49.3	53.4	55.9	58.2	59.0	56.7	51.1	47.3	43.4
		MCDB	45.7	46.7	49.0	53.2	59.2	61.4	64.7	64.7	60.6	53.8	49.2	45.0
	10%	WB	43.4	43.9	45.5	47.8	52.0	54.6	57.0	57.8	55.5	50.0	46.0	42.4
		MCDB	44.7	45.9	47.9	51.4	57.6	59.5	62.8	63.2	59.3	52.5	47.9	44.3
Mean Daily Temperature Range	5% DB	MDBR	4.7	6.5	8.2	10.0	11.7	10.8	12.0	12.1	10.4	6.7	4.9	4.4
		MCDBR	5.2	7.8	10.9	14.7	17.4	16.2	16.0	16.4	14.0	9.8	5.8	5.4
	5% WB	MCWBR	4.1	4.7	5.3	6.1	7.1	6.5	6.4	6.5	6.6	5.4	4.3	4.0
		MCDDBR	4.9	5.8	8.1	12.4	14.9	14.2	14.5	14.7	12.2	8.5	5.3	5.4
Clear-Sky Solar Irradiance	taub		0.312	0.316	0.343	0.362	0.379	0.377	0.364	0.367	0.342	0.366	0.321	0.308
		taud	2.403	2.476	2.418	2.376	2.371	2.415	2.492	2.497	2.557	2.380	2.457	2.370
	Ebn at Noon		229	263	273	280	279	279	282	276	272	237	225	210
		Edh at Noon	20	24	30	35	36	35	32	30	26	26	19	18
All-Sky Solar Radiation	RadAvg		238	508	762	1192	1566	1666	1813	1573	1086	579	278	189
	RadStd		37	92	110	123	175	188	217	135	100	104	56	30

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page